Class Manual 5

1. **Find students enrolled in courses taught by Dr. Smith.**

   QUERY

```
SELECT name
FROM Student
WHERE student_id IN (
    SELECT student_id
    FROM Enrollment
    WHERE course_id IN (
        SELECT course_id
        FROM Course
        WHERE professor_id = (
            SELECT professor_id
            FROM Professor
            WHERE professor_name = 'Dr. Smith'
        )
    )
);
```

   OUTPUT

	name
▶	Aman
	Riya
	Rahul
	Neha
	Sneha

2. **Find courses whose enrollment is greater than minimum enrollment among all four courses.**

   QUERY

```
SELECT course_id, course_name
FROM Course
WHERE course_id IN (
    SELECT course_id
    FROM Enrollment
    GROUP BY course_id
    HAVING COUNT(*) > (
        SELECT MIN(cnt)
```

```
        FROM (
            SELECT COUNT(*) AS cnt
            FROM Enrollment
            GROUP BY course_id
        ) AS temp
    )
);
```

OUTPUT

	course_id	course_name
▶	101	DBMS
	102	OS
	103	Networks

3. Find students whose CGPA is among top 3 highest CGPA

QUERY

```
SELECT name, cgpa
FROM Student
WHERE cgpa IN (
    SELECT cgpa
    FROM (
        SELECT DISTINCT cgpa
        FROM Student
        ORDER BY cgpa DESC
        LIMIT 3
    ) AS temp
);
```

OUTPUT

	name	cgpa
▶	Riya	9.10
	Neha	9.50
	Priya	9.30

4. Display courses that have no students enrolled.

QUERY

```
SELECT course_name
FROM Course
WHERE course_id NOT IN (
```

```
    SELECT DISTINCT course_id
    FROM Enrollment
);
```

OUTPUT

	course_name
▶	ML

5. **Find students who are enrolled in every course offered.**

QUERY

```
SELECT name
FROM Student
WHERE student_id IN (
    SELECT student_id
    FROM Enrollment
    GROUP BY student_id
    HAVING COUNT(DISTINCT course_id) = (
        SELECT COUNT(*) FROM Course
    )
);
```

OUTPUT

	name

No Students

6. **List courses where maximum grade is greater than average grade for all courses.**

QUERY

```
SELECT course_name
FROM Course
WHERE course_id IN (
    SELECT course_id
    FROM Enrollment
    GROUP BY course_id
    HAVING MAX(grade) > (
        SELECT AVG(grade) FROM Enrollment
    )
);
```

OUTPUT

course_name
DBMS
OS
Networks
AI

7. **Find students who are not enrolled in any course.**

QUERY

```
SELECT name
FROM Student
WHERE student_id NOT IN (
    SELECT DISTINCT student_id
    FROM Enrollment
);
```

OUTPUT

name

No students

8. **Display courses having highest number of enrollments.**

QUERY

```
SELECT course_name
FROM Course
WHERE course_id IN (
    SELECT course_id
    FROM Enrollment
    GROUP BY course_id
    HAVING COUNT(*) = (
        SELECT MAX(cnt)
        FROM (
            SELECT COUNT(*) AS cnt
            FROM Enrollment
            GROUP BY course_id
        ) AS temp
    )
);
```

OUTPUT

	course_name
▶	DBMS